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B. E. (Eighth Semester) Examination, April-May, 2016

(New Scheme)

(Civil Engg. Branch)

COMPUTER APPLICATIONS in CIVIL ENGINEERING

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

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Note : *Use C++ programming language to solve all the questions. Assume suitable data where necessary and mention if clearly. Attempt all questios. Part (a) of each question is compulsory in each unit. Attempt any* ***one*** *part from (b) & (c). Algorithm and flowchart is compulsory for part (b) and (c).*

Unit-I

1. (a) Write a program to determine Froude Number. 2

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(b) Write a program to compute friction factor for Laminar flow and turbulent flow. 14

(c) Write a program to compute discharge through open rectangular channel if depth is known? 14

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Unit-II

2. (a) Write an algorithm to convert degrees into radians and radians into degrees. 2

(b) Write a program to Convert Whole Circle Bearing to Quadrantal Bearing 14

(c) Write a program to compute reduced levels using rise and fall method. 14

Unit-III

3. (a) Write an expression for Computing Porosity. 2

(b) Write a program to determine the one dimensional pre consolidated settlement under compacted fill. 14

(c) Write a program to compute total stress, pore water pressure and effective stress under hydrostatic conditions. 14

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Unit-IV

4. (a) Draw the flow chart for computing maximum deflection in a simply supported beam carrying a point load at the mid span. 2

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(b) Write a program to compute bending moment, shear force and deflection at every quarter point in a simply supported beam carrying a uniformely distributed load. 14

(c) Write a program to calculate deflection value at every tenth span carrying uniformly distributed load throughout the span. 14

Unit-V

5. (a) Draw the flow chart for moments of resistance of a balanced section. 2

(b) Write a program to compute the area of steel in an under reinforced section by limit state method, if the factored moment at a section is given. 14

(c) Write a program to compute the permissible stress in bending compression for a laterally unsupported beam of given section.

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